

Decision-Focused Learning for Kidney Exchange Programs

Internship Research Project

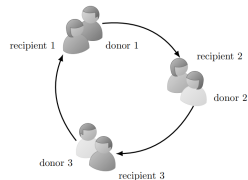
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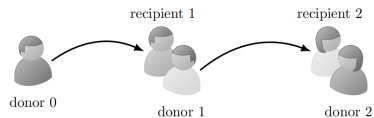
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Why Kidney Exchange Matters

- Many patients need kidney transplants, but compatible living donors are scarce.
- A willing donor may be medically incompatible with the intended recipient.
- Kidney exchange pools incompatible pairs and selects swaps.
- Non-directed donors can start chains that help multiple patients.



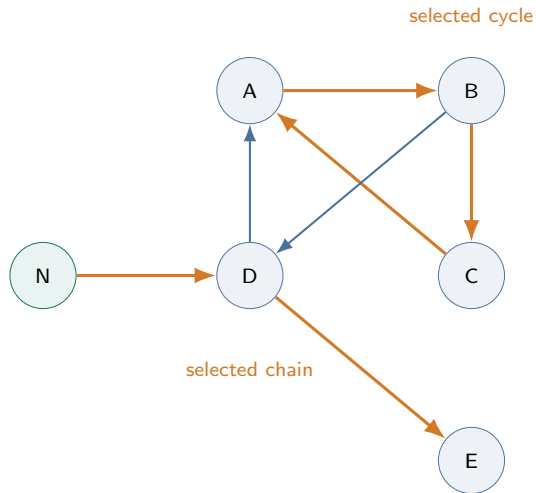
Cycle



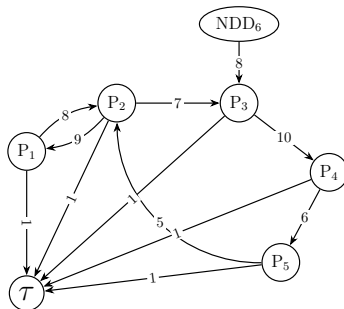
Chain

Kidney Exchange as a Graph Optimization Problem

- **Sets:** $\mathcal{R}$: recipient-donor pairs, $\mathcal{N}$: non-directed donors.
- **Graph:** $G = (\mathcal{V}, \mathcal{A})$, with vertices $\mathcal{V} = \mathcal{R} \cup \mathcal{N}$.
- **Limits:** K : maximum cycle length, L : maximum chain length.
- **Cycle:** $\langle v_1, \dots, v_k, v_1 \rangle$, $k \leq K$, using distinct vertices in $\mathcal{R}$.
- **Chain:** starts from $n \in \mathcal{N}$, then visits compatible pairs, with length at most L .
- **Goal:** select vertex-disjoint cycles and chains maximizing total edge weight w_{uv} .



Example



Instance.

$$\mathcal{R} = \{P_1, P_2, P_3, P_4, P_5\}, \quad \mathcal{N} = \{NDD_6\}, \quad K = 4, \quad L = 4.$$

Possible Cycles (RDP-only closed loops)

- Cycle 1 (2-way): $P_1 \rightarrow P_2 \rightarrow P_1$; length 2.
- Cycle 2 (4-way): $P_2 \rightarrow P_3 \rightarrow P_4 \rightarrow P_5 \rightarrow P_2$; length 4.

Possible Chains (start at NDD, end at τ)

- Chain 1: $NDD_6 \rightarrow P_3 \rightarrow \tau$; length 2.
- Chain 2: $NDD_6 \rightarrow P_3 \rightarrow P_4 \rightarrow \tau$; length 3.
- Chain 3: $NDD_6 \rightarrow P_3 \rightarrow P_4 \rightarrow P_5 \rightarrow \tau$; length 4.
- Chain 4: $NDD_6 \rightarrow P_3 \rightarrow P_4 \rightarrow P_5 \rightarrow P_2 \rightarrow \tau$; length 5.
- Chain 5: $NDD_6 \rightarrow P_3 \rightarrow P_4 \rightarrow P_5 \rightarrow P_2 \rightarrow P_1 \rightarrow \tau$; length 6.

What Makes the Problem Difficult?

Two difficulties:

① Optimization difficulty

- Cycles/chains create a combinatorial problem.
- Bounded cycle length > 2 is NP-hard.

② Prediction difficulty

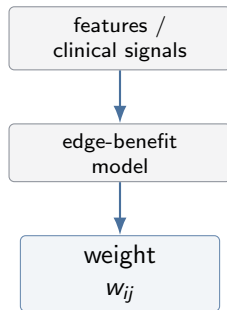
- The benefit of a transplant edge is uncertain.
- A good edge-level prediction may not lead to the best final exchange plan.

From Transplant Benefit to Edge Weights

Each possible transplant edge needs a benefit score.

Clinical benefit components

- transplant success probability
- Quality-Adjusted Life Year (QALY)
- fairness or priority terms
- patient-donor compatibility quality

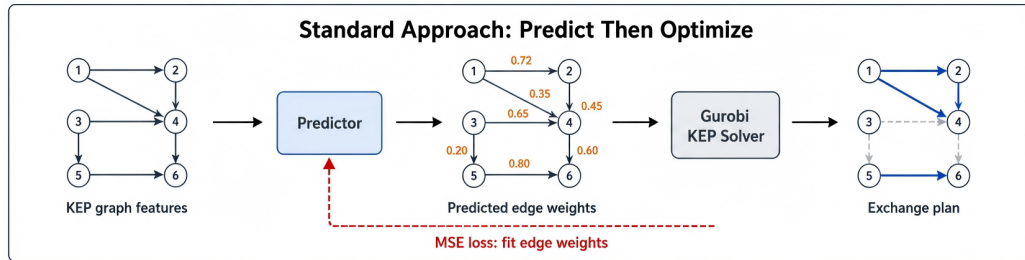


What the optimizer sees

- a weighted compatibility graph
- edge $i \rightarrow j$ has scalar weight w_{ij}
- cycles/chains are selected using these weights

$$i \rightarrow j \rightsquigarrow w_{ij}$$

Standard Approach: Predict Then Optimize



Step 1: train ML model

features $\rightarrow$ predicted edge weights

Step 2: run KEP optimizer

predicted weights $\rightarrow$ selected cycles/chains

Problem

Prediction loss $\neq$ decision quality.

Research Idea: Learning for the Final Exchange Plan

Core idea

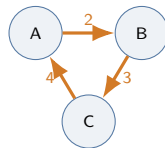
- We care about the final KEP decision gap.
- But the discrete solver is not directly differentiable.
- Train the reward model with a decision-focused surrogate.

Questions

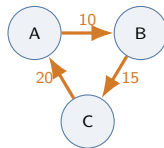
- Can surrogate training reduce the decision gap compared with MSE?
- Does the improvement generalize to held-out or unseen KEP graphs?
- When does the surrogate advantage disappear?

Prediction accuracy $\neq$ decision quality

Predicted weights



Same weights $\times 5$



Goal: learn weights that induce better exchange plans, not necessarily exact edge weights.

Why End-to-End Training Is Hard

The right objective is the final decision gap, but it is not an easy training signal.

Challenge 1: Non-differentiability

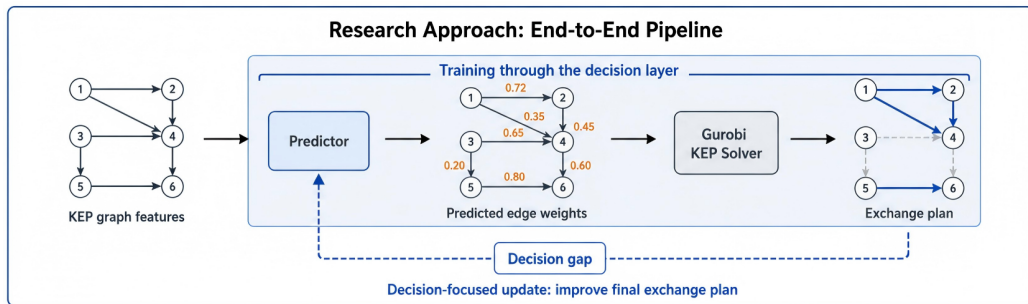
- The KEP solver returns a discrete exchange plan.
- Small changes in predicted weights may not change the selected plan.
- The true decision gap has zero or undefined gradients.

Challenge 2: Computational cost

- Training now calls the KEP solver inside the learning loop.
- Perturbation-based surrogates require many repeated solver calls.
- Full end-to-end training is much more expensive than MSE.

Design response: same graph data + same solver + different training objectives.

Research Approach: Experimental Pipeline



Solutions

1. Surrogate gradients

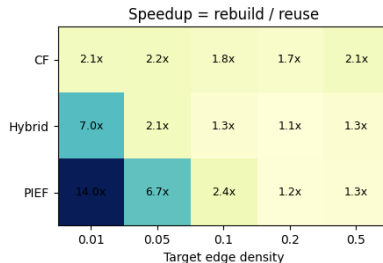
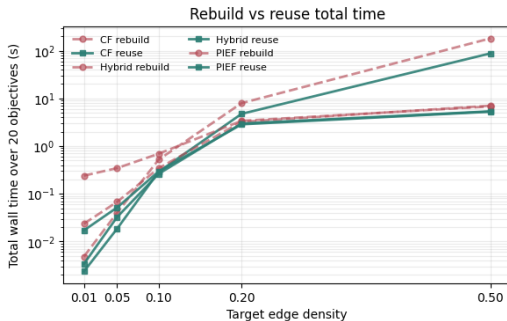
- Use a decision-focused surrogate loss.
- Convert discrete solver outputs into a usable training signal.
- FY perturbation is the main surrogate in this study.

2. Gurobi model reuse

- Reuse cycle/chain enumeration and model structure when possible.
- Update edge weights/objectives instead of rebuilding every solve.
- Reduce the cost of repeated solver-in-the-loop.

Engineering Check: Solver Reuse Benchmark

Reuse benchmark: $n=50$, $K=3$, $L=4$, 20 random objective vectors



Finding

- Reuse is faster in every formulation; gains are largest when model building dominates.
- Dense CF: 181.2s $\rightarrow$ 87.9s; Hybrid/PIEF: about 7s $\rightarrow$ 5.3s.

Experiment Settings

Purpose

Use a controlled two-dimensional setting to compare reward fitting with decision-focused surrogate training.

Reward model

$$\hat{w}_e(\theta) = \theta_1 u_e + \theta_2 c_e$$

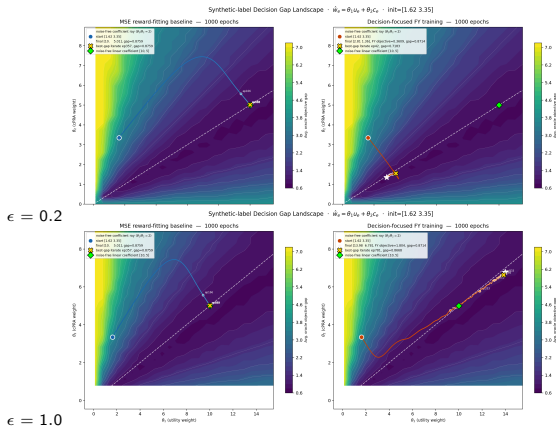
- u_e : edge utility feature.
- c_e : recipient cPRA feature.
- θ_1 : utility weight.
- θ_2 : cPRA weight.

Synthetic label generation

$$w_e^{\text{syn}} = \max\{0, (10u_e + 5c_e)(1 + \xi_e)\}$$

- Noise-free coefficient: $[10, 5]$.
- Multiplicative edge noise: $\sigma = 0.10$.
- Labels are synthetic edge benefits, not clinical outcomes.

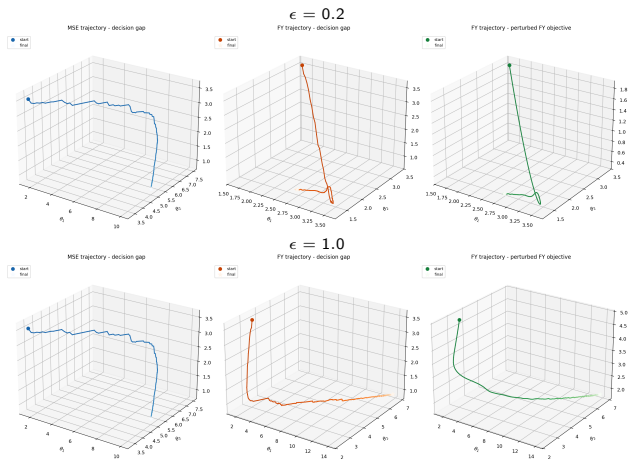
Experiment: Contour Landscapes



Observations

- MSE moves toward the clean-label direction [10, 5].
- FY follows a decision-induced path, not exact coefficient recovery.
- Smaller ϵ gives a lower-scale FY endpoint.
- Both FY runs target decision quality rather than edge-level MSE fit.

Experiment: 3D Trajectories



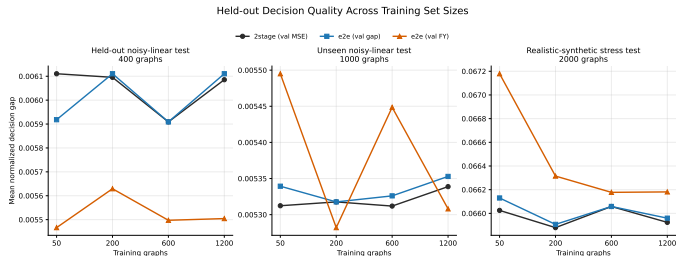
Observations

- 3D plots separate parameter motion from decision-gap and FY-objective changes.
- FY quickly moves toward a favorable region, then may plateau or oscillate.
- Decision gap and FY objective do not always improve monotonically together.
- The best decision-gap checkpoint can occur before the final epoch.

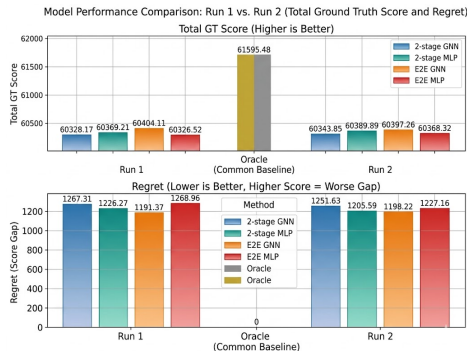
Experiment: Generalization Study

Observations

- Metric: normalized decision gap; lower is better.
- FY-loss-selected E2E improves on the original held-out split.
- The advantage weakens on newly generated unseen graphs.
- Under realistic-synthetic label shift, two-stage is tied or better.
- The surrogate advantage depends on validation and distribution match.



Additional Result: Full MLP/GNN KEP Pipeline



Observations

- Compared two-stage MLP/GNN, E2E MLP/GNN, and an oracle using ground-truth weights.
- E2E GNN gives modest repeatable gains over the two-stage GNN baseline.
- E2E MLP does not consistently improve over two-stage MLP; the oracle gap remains substantial.

Takeaways: Promising, but Not Yet Robust

Decision-focused surrogate learning can improve controlled KEP decisions, but the gain is not automatic.

What worked

- DFL changes the learned reward geometry.
- It can reduce downstream decision gap in controlled settings.
- It optimizes for final plans, not only edge prediction.

Limitations

- Synthetic labels, not clinical outcomes.
- Simple two-feature Step1 probe.
- Limited seeds and distribution settings.
- Solver-in-the-loop training is expensive.

Next steps

- Richer clinical edge-benefit models.
- Stronger GNN or structured predictors.
- Larger unseen tests and multiple seeds.
- Better surrogate validation strategy.

Conclusion: useful signal in controlled benchmarks; robustness remains the key challenge.